

Self-Supervised I-JEPA for Label-Efficient Pneumonia Screening from Chest X-Rays in Resource-Limited Settings

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Abstract. Pneumonia causes over 2.2 million deaths annually, yet accurate chest X-ray (CXR) interpretation remains inaccessible in resource-limited settings due to radiologist shortages and scarce annotated data. This study applies I-JEPA to binary pneumonia classification by pretraining a ViT-Small/16 encoder on 50,000 unlabeled NIH ChestX-ray14 images via random patch masking, then fine-tuning on the RSNA Pneumonia Detection Dataset. Unlike pixel-reconstruction methods, I-JEPA predicts masked-region latent representations, compelling the encoder toward high-level anatomical features relevant to diagnosis. Full fine-tuning achieves AUC = 0.8297 and Recall = 0.7738, outperforming ImageNet baselines in recall across all thresholds; at 1% labeled data, I-JEPA attains AUC = 0.8041 ± 0.0002 , surpassing baselines with 143-fold lower variance. CLAHE preprocessing uniformly degrades performance (Δ AUC = -0.047), revealing an undocumented SSL-preprocessing interaction. These findings establish I-JEPA as a practical, annotation-efficient framework for pneumonia screening in resource-limited settings.

Keywords: *Self-Supervised Learning · Chest X-Ray · Pneumonia Classification · I-JEPA · Label Efficiency*

1 Introduction

Pneumonia remains one of the most significant infectious causes of morbidity and mortality worldwide. The World Health Organization attributes approximately 2.2 million deaths to pneumonia in 2021, including more than 502,000 children under five [1]. Chest X-ray (CXR) imaging is the most widely used diagnostic modality due to its low cost and broad availability, but accurate interpretation requires specialized radiological expertise that is not uniformly accessible. In many developing countries, including Vietnam, district hospitals often lack permanently stationed radiologists, contributing to delayed diagnoses and increased interpretation errors.

Deep learning offers a path toward automated decision support. CNNs and Vision Transformers (ViTs) have achieved competitive performance on thoracic disease classification [2,3], but most approaches rely on supervised learning requiring large quantities of expert-annotated data, a major bottleneck in medical imaging. Self-supervised learning (SSL) addresses this by leveraging unlabeled images to learn visual representations prior to downstream fine-tuning; domain-specific SSL pretraining has been shown to outperform ImageNet-supervised initialization when labels are scarce [4]. Among recent SSL approaches, I-JEPA [5] predicts latent representations of masked regions rather than reconstructing pixels

(as in MAE [6]) or relying on augmentation-based contrastive objectives (SimCLR [7], BYOL [8]). This predictive objective encourages the model to capture high-level semantic structure, a property well-suited to medical imaging, where diagnosis depends on anatomical patterns rather than low-level appearance. The effectiveness of I-JEPA for pneumonia classification from CXR remains largely unexplored.

This study investigates three research questions: (RQ1) what distinguishing characteristics in-domain I-JEPA pretraining confers relative to ImageNet-supervised pretraining when the encoder architecture is held constant; (RQ2) whether I-JEPA exhibits superior label efficiency across labeled-data regimes from 1% to 100%; and (RQ3) whether regions emphasized by the fine-tuned model correspond to radiologist-annotated lesions. Three contributions follow: (1) a systematic evaluation of I-JEPA with random patch masking for pneumonia classification on the RSNA benchmark; (2) a label-efficiency analysis comparing I-JEPA against CNN and Transformer baselines across six labeled-data fractions; and (3) a quantitative explainability study using Grad-CAM and Attention Rollout against radiologist-provided lesion annotations.

2 Related Work

2.1 Chest X-Ray Classification and Self-Supervised Learning

Before the emergence of self-supervised approaches, supervised CNN architectures dominated automated pneumonia detection from CXR. Rajpurkar et al. [9] introduced CheXNet, a 121-layer DenseNet trained on over 100,000 frontal-view radiographs, reporting radiologist-level performance on pneumonia detection as measured by F1-score; this result established deep CNNs as clinically credible tools for thoracic disease screening and motivated subsequent benchmark efforts. Kundu et al. [10] later demonstrated that ensembling multiple CNN backbones (GoogLeNet, ResNet-18, DenseNet-121) via weighted averaging improves robustness over any single architecture on pneumonia classification, achieving accuracy gains attributable to complementary error patterns across networks. Both lines of work, however, share the same fundamental limitation that motivates the present study: their reported performance presupposes access to large labeled training corpora, a condition rarely satisfied in primary-care deployment settings.

Wang et al. [2] established the foundational CXR benchmark via NIH ChestX-ray14 (>112,000 images, 14 NLP-derived labels), though label accuracy is estimated at only $\approx 90\%$ and pneumonia constitutes just 1.3% of samples. RSNA [11] subsequently provided $\approx 30,000$ expert-verified binary labels with bounding boxes, becoming the de facto pneumonia detection standard. In the Vietnamese context, Khiem et al. [12] demonstrated feasibility of supervised deep learning for pulmonary disease classification, underscoring the persistent challenge of label scarcity at the primary-care level.

Within SSL, MoCo [13] and SimCLR [7] rely on contrastive objectives that risk pushing similar pathological images apart and require large GPU memory. Sowrirajan et al. [14] applied MoCo specifically to chest radiographs (MoCo-CXR), showing that contrastive pretraining on unlabeled CXR data improves both classification AUC and transfer robustness relative to ImageNet initialization, an early demonstration of the in-domain pretraining advantage that

motivates the I-JEPA-based approach adopted here, though MoCo-CXR remains constrained by the same false-negative risk inherent to contrastive objectives in medical imaging. BYOL [8] eliminated negative pairs via an EMA-updated target network, a mechanism later inherited by I-JEPA. MAE [6] instead reconstructs masked pixels, forcing the encoder toward low-level surface detail rather than clinically relevant semantics. Azizi et al. [4] provided the first systematic evidence that in-domain SSL pretraining outperforms ImageNet baselines in the low-label regime, motivating the present study. I-JEPA [5] resolves prior limitations by predicting directly in latent space; cross-domain adaptations, including A-JEPA [15] and Audio-JEPA [16] for audio, Brain-JEPA [18] for fMRI, and RadJEPA [19] for CXR, confirm the flexibility of JEPA's masking strategy across domains. CrossJEPA [17] further demonstrates that the predictive objective itself, rather than masking specifically, is the core transferable mechanism, achieving state-of-the-art 3D representation learning from 2D images without any masking step.

2.2 Research Gaps

Three gaps remain unaddressed: (1) no study has evaluated I-JEPA for binary pneumonia classification on RSNA against CNN/Transformer baselines sharing an identical encoder; (2) no granular label-efficiency analysis exists for I-JEPA versus supervised baselines on CXR data; (3) no quantitative explainability evaluation has used radiologist bounding boxes as ground truth for I-JEPA on CXR. This study addresses all three.

3 Materials and Methods

3.1 Datasets and Preprocessing

Two public datasets are used: NIH ChestX-ray14 [2], from which 50,000 images are randomly sampled as an unlabeled pretraining source (the 14 NLP-derived labels are excluded due to noise and pneumonia under-representation), and the RSNA Pneumonia Detection Dataset [11], providing $\approx 30,000$ expert-labeled images with bounding boxes for fine-tuning and evaluation. Zero patient-level overlap between the two datasets is confirmed. RSNA is partitioned via stratified splitting (seed = 42) into train (18,678), validation (4,003), and test (4,003) sets, each maintaining a 22.5% positive rate; class imbalance ($\approx 1:3.44$) is addressed via `pos_weight` in `BCEWithLogitsLoss`.

DICOM images are converted to 8-bit PNG, replicated to 3-channel, resized to 224×224 , and normalized using ImageNet statistics. Mild augmentation (horizontal flip $p=0.5$, rotation $\pm 7^\circ$, brightness jitter ± 0.2 , translation $\leq 5\%$) is applied only to the RSNA training split during fine-tuning, excluding vertical flip and aggressive cropping to preserve diagnostic spatial structure. CLAHE (`clip_limit=2.0`, `tile=8 \times 8`) is evaluated separately as an ablation.

3.2 Proposed I-JEPA Framework

The framework adapts I-JEPA [5] with random patch masking for CXR pretraining. The Student Encoder is a ViT-Small/16 ($d=384$, 12 blocks, 6 heads) processing all 196 patch tokens (16×16 patches, 224×224 input). The Target Encoder is an architectural copy whose parameters

θ' are updated exclusively through an Exponential Moving Average (EMA) of the Student Encoder's parameters θ , with no gradient flowing into θ' directly:

$$\theta' \leftarrow m \cdot \theta' + (1 - m) \cdot \theta \quad (1)$$

where the momentum coefficient m is increased linearly from 0.996 at epoch 1 to 0.999996 at epoch 50, following the schedule of the original I-JEPA publication [5]. The Predictor is a narrow 4-layer, 4-head Transformer (embedding dim 256) that receives the 196 Student embeddings and predicts embeddings for 147 randomly masked target positions (mask ratio = 0.75).

Random masking, rather than the original block masking, was adopted for two reasons: (1) cross-domain evidence that random masking achieves state-of-the-art results when JEPA is transferred to new domains [15,16,19]; and (2) the high bilateral symmetry of frontal CXR images, which favors uniform anatomical coverage. The pretraining objective is the mean-squared error computed in latent space between the Predictor's output and the Target Encoder's embedding at each masked position i in the target set T ($|T| = 147$):

$$L = (1/|T|) \cdot \sum_{i \in T} \|g\phi(f\theta(x))_i - f\theta'(x)_i\|^2 \quad (2)$$

where $f\theta$ is the Student Encoder, $f\theta'$ is the Target Encoder, $g\phi$ is the Predictor, and x is the input image. No pixel reconstruction is performed at any stage; the Target Encoder receives no gradient, consistent with Eq. (1).

The EMA update rule serves a purpose beyond mere stabilization: it directly prevents representational collapse, a well-documented failure mode of joint-embedding architectures in which the encoder learns to output a constant vector regardless of input, trivially minimizing the loss without acquiring useful structure. If both encoders were updated synchronously via shared gradients, this collapse would be the path of least resistance. By contrast, the asymmetric EMA update ensures the Target Encoder always lags slightly behind the Student Encoder, producing a target distribution that drifts slowly enough to remain learnable yet is never directly optimized by the loss it provides, a property formally analyzed in BYOL [8] and inherited unchanged by I-JEPA. The momentum schedule used here (0.996 to 0.999996) follows the original I-JEPA configuration [5] exactly, since the schedule's role is architectural (preventing collapse) rather than dataset-specific, and no evidence in the cross-domain JEPA literature [15,16,19] suggests this schedule requires retuning for new imaging domains.

Following pretraining, the encoder is augmented with a binary classification head (Linear(384,1)) and fine-tuned under five configurations: Linear Probing (LP), Partial Fine-Tuning (Partial FT), and Full Fine-Tuning (Full FT), each in two variants: v1 (fixed learning rates) and v2 (Layer-wise Learning Rate Decay, LLRD). Key hyperparameters are summarized in Table 1.

Table 1. Key fine-tuning hyperparameters by strategy.

Strategy	Hyperparameter	Config v1	Config v2 (LLRD)
Linear Probe	Head LR / Epochs	1×10^{-3} / 15	1×10^{-3} / 15

Partial FT	Head / Enc. LR	$1 \times 10^{-4} / 1 \times 10^{-5}$	$3 \times 10^{-5} / 5 \times 10^{-6} + \text{LLRD}$
Full FT	Head / Enc. LR	$5 \times 10^{-5} / 5 \times 10^{-6}$	$3 \times 10^{-5} / 1 \times 10^{-6} + \text{LLRD}$
Shared	Dropout / Wt. decay	0.2 / 0.05	0.1 / 0.05

3.3 Baselines and Experimental Setup

Two supervised baselines provide controlled comparison: ResNet50 [20] (IMAGENET1K_V2 weights, $\sim 25\text{M}$ params) and ViT-Small/16 [3] (DeiT ImageNet-21k $\rightarrow$ 1k weights via the timm library [21], $\sim 22\text{M}$ params), the latter sharing the identical backbone as the I-JEPA encoder to isolate the effect of pretraining method from architecture. Both baselines are retrained from scratch under identical splits, preprocessing, loss, and evaluation protocol as I-JEPA, eliminating confounding variables across studies.

All experiments use PyTorch 2.x with timm [21] and pydicom, on Kaggle Notebooks with dual NVIDIA T4 GPUs (16GB each). I-JEPA pretraining ran 50 epochs on 50,000 NIH images (AdamW, peak LR 1×10^{-4} , batch size 32×2 accumulation, FP16, mask ratio 0.75), with MSE loss decreasing from 0.1185 to 0.0041 (96.5% reduction) without mode collapse; pretraining required ≈ 4 hours, fine-tuning ≈ 6 hours. Label efficiency is assessed across six labeled-data fractions (1–100%); explainability uses Pointing Game Accuracy and Overlap Ratio against radiologist bounding boxes.

Classification performance is evaluated primarily via AUC-ROC, given the class imbalance in RSNA (22.5% positive rate). Letting TP, TN, FP, and FN denote true/false positives and negatives with pneumonia as the positive class, Recall (Sensitivity), Specificity, and F1-score are defined as:

$$\text{Recall} = TP / (TP + FN) \quad \text{Specificity} = TN / (TN + FP) \quad (3)$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}), \quad \text{Precision} = TP / (TP + FP) \quad (4)$$

Decision thresholds are additionally selected via Youden's J statistic, which identifies the operating point maximizing the unweighted balance between sensitivity and specificity:

$$J = \text{Recall} + \text{Specificity} - 1 \quad (5)$$

Eq. (5) is reported alongside F1-optimal and recall-targeted thresholds in Sect. 4.3, since no single criterion is clinically sufficient on its own: J treats sensitivity and specificity symmetrically, whereas the screening context motivating this study assigns asymmetric cost to false negatives (Sect. 4.1), making F1- or recall-targeted thresholds more clinically relevant than the J-optimal point in isolation.

4 Results

4.1 Baseline and I-JEPA Fine-Tuning Performance

Table 2 reports baseline and I-JEPA fine-tuning results on the RSNA test set (4,003 images; 902 positive). ResNet50 achieves the highest AUC (0.8862) but only Recall = 0.5421 at the

default threshold, missing nearly 46% of true pneumonia cases. ViT-Small/16 ImageNet yields comparable AUC (0.8797) but exhibits a substantial domain-induced shift in its prediction distribution: Recall collapses to 0.2783 at threshold 0.50, requiring an optimal threshold of 0.24, a direct indicator of the ImageNet-to-CXR domain gap. I-JEPA Full FT v1 achieves AUC = 0.8297 with Recall = 0.7738 at the default threshold, approximately $2.77\times$ higher than the architecturally identical ViT-ImageNet baseline, demonstrating that in-domain self-supervised pretraining shifts the model's prediction distribution toward higher sensitivity independently of peak AUC.

Table 2. Model performance on the RSNA test set (threshold = 0.50 unless noted). Best Thr. is the post-hoc F1-optimal threshold identified on this same test set for descriptive purposes (see Sect. 4.3); it is not used for model selection and does not affect the AUC values reported.

Model	AUC	Recall	Specificity	F1@0.50	F1 (best-thr)	Best Thr.
ResNet50 ImageNet	0.8862	0.5421	0.9439	0.6249	0.6508	0.37
ViT-Small/16 ImageNet	0.8797	0.2783	0.9836	0.4169	0.6379	0.24
I-JEPA Linear Probe	0.7737	0.6918	0.7362	0.5324	0.5344	0.49
I-JEPA Partial FT v1	0.8003	0.7417	0.7291	0.5550	0.5630	0.56
I-JEPA Partial FT v2	0.8011	0.7029	0.7552	0.5525	0.5596	0.55
I-JEPA Full FT v1	0.8297	0.7738	0.7333	0.5752	0.5921	0.60
I-JEPA Full FT v2 (LLRD)	0.8041	0.7195	0.7462	0.5552	0.5661	0.55

Note: F1@0.50 denotes F1-score at the default threshold (0.50), the basis for all cross-model comparisons in this study. F1 (best-thr) denotes F1 at the post-hoc operating point in the rightmost column, reported as a descriptive upper bound rather than a model-selection criterion, since threshold is applied after training and does not affect AUC.

Classification performance improves monotonically as more encoder layers are unfrozen (AUC: 0.7737 $\rightarrow$ 0.80 $\rightarrow$ 0.8297), confirming that task-specific adaptation is necessary while the non-trivial Linear Probe AUC validates pretrained representation quality. Full FT v2 (LLRD) underperforms v1, likely due to the absence of the sequential fine-tuning initialization available to v1. Contextualizing against the closest comparable published result, Yerzhanuly et al. [22] (preprint) report ResNet-50 AUC = 0.8745 on a class-balanced 12,000-image RSNA subset; the present ResNet-50 baseline (AUC = 0.8862, full 26,684-image dataset at realistic 22.5% prevalence) is competitive, supporting the validity of the experimental setup. I-JEPA's AUC = 0.8297 reflects a constrained budget (50 epochs vs. 300–800 in the original publication) and should be read as a lower bound; its primary contribution is not peak AUC but a more favorable decision threshold behavior and variance reduction under label scarcity, detailed next.

Examining the confusion matrices underlying Table 2 clarifies the clinical character of each model's errors. At the default threshold, ResNet50 produces 413 false negatives out of 902 true pneumonia cases (46% miss rate) but only 174 false positives out of 3,101 normal cases, an error profile favoring specificity at the direct expense of sensitivity, which is undesirable for

a screening tool whose primary clinical role is to avoid missed diagnoses. I-JEPA Full FT v1 inverts this trade-off, producing 204 false negatives (23% miss rate) against 827 false positives (27% false-alarm rate). In a primary-care screening context, this inverted error profile is clinically preferable: a false positive triggers a confirmatory chest reading by an available clinician, whereas a false negative delays treatment of an active infection with no compensating safeguard. The practical cost asymmetry between these two error types, not captured by AUC, which weights both equally, is precisely why Recall and the threshold-tuning analysis in Sect. 4.3 are emphasized as the clinically actionable metrics in this study, rather than AUC alone.

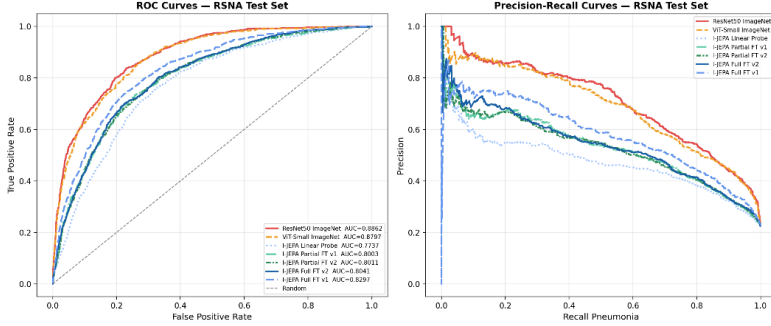


Fig. 1. ROC curves (left) and Precision-Recall curves (right) for all models on the RSNA test set.

4.2 Label Efficiency Analysis (RQ2)

All models were retrained at six labeled-data fractions; at 1% and 5%, experiments were repeated across multiple seeds (5 and 3, respectively) to quantify variance (Table 3, Fig. 2).

Table 3. Label-efficiency AUC across six labeled-data fractions (mean $\pm$ std for 1–5%; single run for $\geq 10\%$). ViT-Small/16 was not evaluated at 10–50% due to compute constraints.

Labels	n images	I-JEPA Partial v1	I-JEPA Full v2	ResNet50	ViT-Small/16
1%	187	0.8004$\pm$0.0003	0.8041$\pm$0.0002	0.7821 $\pm$ 0.0286	0.7906 $\pm$ 0.0250
5%	934	0.8001 $\pm$ 0.0001	0.8041 $\pm$ 0.0004	0.8170 $\pm$ 0.0160	0.8185 $\pm$ 0.0215
10%	1,868	0.7895	0.7830	0.8354	N/A
25%	4,670	0.7983	0.7955	0.8595	N/A
50%	9,339	0.7976	0.7997	0.8591	N/A
100%	18,678	0.8098	0.8137	0.8685	0.8797

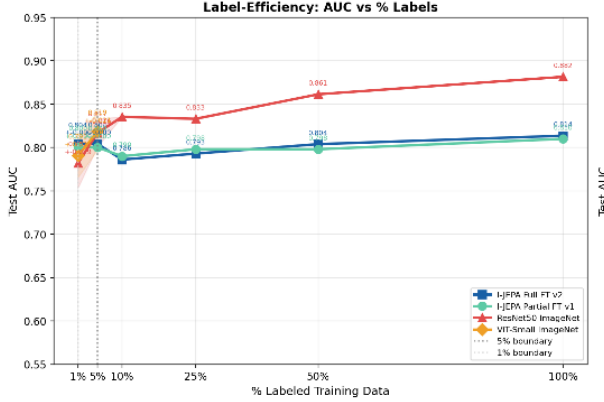


Fig. 2. Label-efficiency crossover: I-JEPA dominates at $\leq 5\%$ labels; supervised baselines recover advantage beyond 10%.

At 1% labeled data, I-JEPA Full FT v2 achieves $\text{AUC} = 0.8041 \pm 0.0002$ versus ResNet50's 0.7821 ± 0.0286 , a 0.022-point AUC improvement accompanied by a 143-fold reduction in performance variance. This stability is arguably more clinically significant than the AUC gain itself: a facility deploying a model whose performance ranges from 0.754 to 0.811 depending on random seed cannot reliably anticipate its behavior, which is incompatible with clinical safety requirements.

An apparent drop at the 5%→10% boundary (e.g., Full v2: 0.8041→0.7830) reflects a deliberate fine-tuning regime change rather than true degradation: at $\leq 5\%$ labels the encoder is fully frozen to prevent overfitting, while from 10% onward it is partially unfrozen for task-specific adaptation. A supplementary frozen-encoder run at 10% yielded $\text{AUC} = 0.8021$, confirming the drop is attributable entirely to the regime change. A genuine performance crossover nonetheless occurs between 5% and 10% labels: as annotation budget increases, supervised baselines recover their discriminative advantage, reaching 0.8685–0.8797 at 100% versus I-JEPA's 0.8137. Notably, I-JEPA at 1% labels ($\text{AUC} = 0.8041$) approximates ResNet50 at 10% labels (0.8354) more closely than at 100% (0.8685), suggesting in-domain SSL pretraining compresses the annotation requirement by roughly one order of magnitude for comparable performance. This collectively answers RQ2: I-JEPA's label-efficiency advantage is concentrated in the extreme low-label regime ($\leq 5\%$) and is most pronounced at 1%. It is noted that fractions $\geq 10\%$ used a single seed due to compute constraints; the exact crossover location should therefore be treated as approximate pending multi-seed confirmation, though the directional finding is robust to this uncertainty.

4.3 Decision Threshold and CLAHE Ablation

Table 4 illustrates how different operating points reshape the Recall–Specificity trade-off for I-JEPA Full FT v1 ($\text{AUC} = 0.8297$, invariant to threshold choice). These operating points are presented for illustrative deployment planning rather than as a model-selection criterion: since the decision threshold is a post-hoc parameter applied after the model is fixed, rather than a quantity learned from data, exploring its range on the evaluation set does not affect the trained model's parameters or its AUC. For deployment as a primary-care screening tool, the Recall

≥ 0.85 operating point (threshold = 0.39) is recommended, identifying 85.6% of true pneumonia cases at Specificity = 0.640, an appropriate trade-off where the clinical cost of a missed diagnosis exceeds that of a false referral, especially as all flagged cases undergo clinician confirmation.

Table 4. Decision threshold analysis for I-JEPA Full FT v1.

Strategy	Threshold	Recall	Specificity	Application
Default	0.50	0.7738	0.7333	Research benchmarking
Best F1	0.60	0.6896	0.8139	Balanced precision–recall
Recall ≥ 0.85	0.39	0.8559	0.6404	Recommended: district triage
Recall ≥ 0.90	0.28	0.9024	0.5372	High-sensitivity triage only

All configurations were retrained with CLAHE preprocessing applied to both training and evaluation images (Table 5). CLAHE uniformly degraded performance (mean $\Delta\text{AUC} = -0.047$, range -0.015 to -0.069), with I-JEPA configurations consistently more affected than supervised baselines, most severely the frozen-encoder Linear Probe. This is explained by a distribution shift: the encoder was pretrained on NIH images without CLAHE, and introducing CLAHE during fine-tuning disrupts the feature correspondence learned during pretraining. The practical guideline is that preprocessing must remain consistent between pretraining and fine-tuning stages to preserve learned domain-specific representations.

Table 5. CLAHE ablation ($\Delta\text{AUC} = \text{with CLAHE} - \text{without CLAHE}$).

Model	ΔAUC	ΔRecall	Interpretation
I-JEPA Linear Probe	-0.069	-0.407	Severe domain mismatch
I-JEPA Full FT v1	-0.052	$+0.008$	AUC drops; Recall unchanged
ResNet50 ImageNet	-0.022	-0.202	Moderate; pretrain already distant
ViT-Small/16 ImageNet	-0.015	-0.196	Mild; domain-agnostic pretrain

4.4 Explainability Analysis (RQ3)

Grad-CAM and Attention Rollout are fundamentally different attribution mechanisms: Grad-CAM [23] uses gradient flow through convolutional features, while Attention Rollout traces Transformer attention propagation and tends to produce more distributed maps. Quantitative comparisons (Table 6) should therefore be interpreted cautiously, as differences may partly reflect XAI method choice rather than true model behavior. Grad-CAM was applied to ResNet50; Attention Rollout was applied to ViT-Small/16 ImageNet and I-JEPA Full FT v1, propagated across all 12 Transformer layers and evaluated against radiologist bounding boxes.

Table 6. Quantitative explainability results (random-baseline Pointing Game $\approx 22\%$).

Model	XAI Method	Pointing Game	Overlap Ratio
ResNet50	Grad-CAM	60.0%	16.6%

ViT-Small/16 ImageNet	Attention Rollout	64.0%	6.8%
I-JEPA Full FT v1	Attention Rollout	48.0%	12.6%

I-JEPA's Pointing Game score of 48.0% more than doubles the $\sim 22\%$ random baseline, providing direct evidence that its attention is not arbitrarily distributed but concentrates meaningfully on radiologist-annotated lesion regions. All three models substantially exceed this random baseline, confirming model attention correlates meaningfully with annotated lesion locations across architectures. ResNet50 achieves the highest Overlap Ratio, while ViT-ImageNet's high Pointing Game score but low Overlap (6.8%) suggests sharply focused attention on a single discriminative sub-region rather than distributed lesion coverage. I-JEPA's comparatively lower Pointing Game score relative to the other two models is attributable to two factors: Attention Rollout is a less direct relevance proxy than Grad-CAM, and I-JEPA pretraining encourages comprehensive global anatomical representations, yielding more distributed rather than focal attention. This provides a positive, albeit partial, answer to RQ3: I-JEPA attention maps show statistically meaningful alignment with expert-annotated lesions, though localization precision relative to other models is constrained by the choice of explainability method, a limitation to be addressed via ViT-specific Grad-CAM in future work.

5 Discussion

The most clinically significant finding is not I-JEPA's absolute AUC but its decision threshold behavior and stability under limited annotation. ViT-ImageNet, despite AUC = 0.8797, requires threshold = 0.24 for competitive sensitivity, a value highly sensitive to deployment-time distributional shift. I-JEPA Full FT v1 instead operates near the natural threshold of 0.50, reflecting a prediction distribution better aligned with the default decision boundary as a result of in-domain pretraining. The 143-fold variance reduction at 1% labeled data is arguably the most practically important result: a model with AUC std of ± 0.0286 may score anywhere between 0.75 and 0.81 depending on seed, too wide a range for reliable clinical specification, whereas I-JEPA's ± 0.0002 provides a tight, predictable bound enabling systematic clinical governance.

It should be made explicit that the Recall comparison in Table 2 is reported at the fixed default threshold of 0.50, not the natural operating point of ResNet50 or ViT-ImageNet (best-F1 thresholds 0.37 and 0.24, respectively), reflecting that these ImageNet-pretrained models were never optimized around 0.50 as a meaningful decision boundary. This is not a cherry-picked comparison: Table 4 confirms that even at I-JEPA's own best-F1 threshold (0.60), its Recall remains substantial (0.6896); the central claim is not that I-JEPA wins at an arbitrarily chosen point, but that its predictions are usable near the conventional default threshold without requiring per-model threshold tuning.

CLAHE's consistently negative effect (mean $\Delta\text{AUC} = -0.047$) contradicts the common assumption that contrast enhancement uniformly improves CXR analysis, revealing an undocumented SSL-preprocessing interaction with direct practical implications: preprocessing must be held fixed across pretraining and fine-tuning. These label-efficiency findings broadly align with Azizi et al. [4] and extend this evidence to the I-JEPA family by explicitly quantifying variance reduction. The present results also extend the contrastive-pretraining

evidence of Sowrirajan et al. [14] (MoCo-CXR) to the predictive paradigm: where MoCo-CXR demonstrated that in-domain contrastive pretraining improves CXR classification over ImageNet initialization, the present findings suggest that predictive latent-space objectives confer a comparable or stronger advantage specifically in the extreme low-label regime, without requiring the large negative-sample batches that contrastive methods demand. RadJEPA [19] demonstrates JEPA at scale achieves state-of-the-art CXR results; this study complements that work by showing meaningful benefits are attainable at substantially smaller scale (50,000 images, 50 epochs, single Kaggle T4), representative of resource constraints at academic medical centers in developing countries.

Several limitations warrant acknowledgement. First, the 50-epoch pretraining budget is substantially shorter than the original publication (300–800 epochs); while the loss converged stably within this budget, extended pretraining may yield additional gains, so results should be read as a lower bound. Second, random masking replaced block masking as a deliberate domain adaptation consistent with A-JEPA, Audio-JEPA, and RadJEPA [15,16,19]; a direct ablation comparing the two masking strategies on CXR data is identified as future work. Third, all data originates from US clinical centers; generalization to Vietnamese populations requires external validation on domain-representative data such as VinDr-CXR [24], the highest-priority direction given this study's clinical motivation. Fourth, the explainability comparison across XAI methods (Grad-CAM vs. Attention Rollout) is not fully controlled; a unified ViT-compatible method (e.g., GradCAM++) is identified as a methodological improvement. Fifth, all experiments used a single hardware configuration (dual Kaggle T4 GPUs); reproducibility across different hardware or software versions was not assessed. Sixth, the AUC gap between I-JEPA and ResNet50 (0.8297 vs. 0.8862) has not been confirmed via DeLong's test or bootstrap confidence intervals, consistent with the single-seed limitation already noted for label fractions $\geq 10\%$ in Sect. 4.2; statistical significance testing for this gap is identified as a priority for future work.

6 Conclusion

This study proposes and evaluates a hybrid self-supervised framework adapting I-JEPA to chest radiograph representation learning via random patch masking, pretrained on 50,000 unlabeled NIH images and fine-tuned on RSNA. Three principal conclusions emerge. First, in-domain self-supervised pretraining substantially improves decision threshold behavior: I-JEPA Full FT v1 achieves Recall = 0.7738 at threshold 0.50, $2.77\times$ higher than the architecturally identical ViT-ImageNet baseline. Second, I-JEPA provides decisive advantages under extreme label scarcity: at 1% labeled data, I-JEPA attains AUC = 0.8041 ± 0.0002 with 143-fold lower variance than ResNet50, approximating ResNet50 performance at $10\times$ the annotation budget. Third, preprocessing consistency between pretraining and fine-tuning is essential, as CLAHE uniformly degrades performance. These findings position in-domain I-JEPA pretraining as a practical, scientifically grounded approach for label-efficient pneumonia screening in primary healthcare settings. Future work includes scaling pretraining to the full NIH corpus, implementing original block masking for direct comparison, applying ViT-specific Grad-CAM, and external validation on VinDr-CXR.

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